Lösungen · Trigonometrie

1. $\mathbf{a = \frac{7{,}4}{\sin 52°} \approx 9{,}4\ m}$ ..... $\sin 52° = \frac{7{,}4}{a}$

2. **a)** $\mathbf{AB \approx 1228{,}7\ m}$ ..... $\sin 55° = AB : 1500$

b) $\mathbf{AD \approx 860{,}4\ m}$ ..... $\cos 55° = AD : 1500$

3. $\mathbf{PB = \frac{1{,}20}{\tan 34{,}9°} \approx 1{,}72\ m}$ ..... $\tan 34{,}9° = 1{,}20 : PB$

4. $\mathbf{AC = \frac{h}{\sin 22°} \approx 78{,}1\ cm}$ ..... $h = \sqrt{32^2 - 13^2} = 3\sqrt{95}$; $\sin 22° = h : AC$

5. $\mathbf{BF \approx 93{,}0\ cm}$ ..... $\sin 45° = BF : 131{,}5$
$\mathbf{BC = BF : \cos 65° \approx 220{,}0\ cm}$ ..... $\cos 65° = BF : BC$

6. $\mathbf{Berg = 57{,}9\ m;\ D = 69{,}9\ m}$ ..... $\sin 30° = BC : 112{,}8$; $Berg = BC + 1{,}5$; $D = 57{,}9 + 12{,}0$

7. $\mathbf{BD \approx 502\ m}$ ..... $\sin 54° = AD : 920$; $\tan 34° = BD : AD$

8. **100 und 6**
ja ..... $\frac{16}{170} = \frac{8}{85} \approx 9{,}4\ \% > 6\ \%$